Day 27 - [8.1.2 | 8.1.3] Polynomial Investigation

Friday, April 21, 2023 12:34 PM

Objective:
- Students will investigate properties of Polynomial equations
- Students will sketch polynomials based on the patterns they made conjectures on last class (zeros, degree, sign of LC, end behavior, and multiplicities)

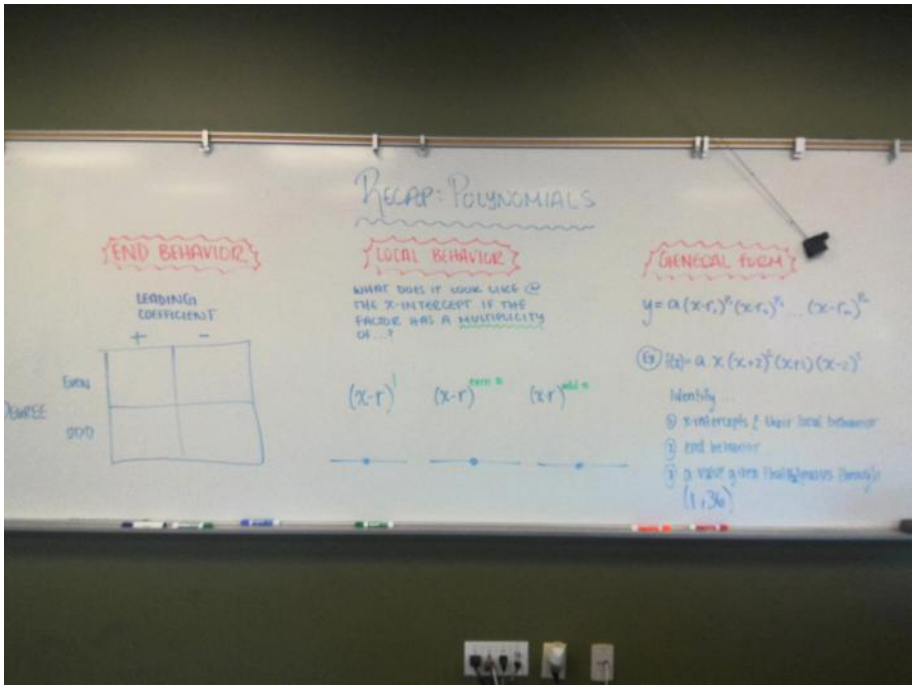
Mathematical Practice(s): reason abstractly and quantitatively, attend to precision, look for and make use of structure

Classwork:
- Review Growth Check
- Pass Back KC1 - Trig
- Warm-Up: Behavior of a Polynomial
- Desmos: Graphs to Equations
  - https://teacher.desmos.com/activitybuilder/custom/609212155cc34d4f3c31054b?utm_campaign=share&utm_content=activity
  - End at Slide 16
- Exit Ticket: Describing Polynomials and Sketching
- **P1:** Complete the Survey

Homework:
- STUDY FOR KC #2 - TRIG!

Announcement:
- Day 29: Trigonometry KC #2
  ODD - Wednesday, April 15th
  EVEN - Thursday, April 16th

Warm-Up



Exit Ticket

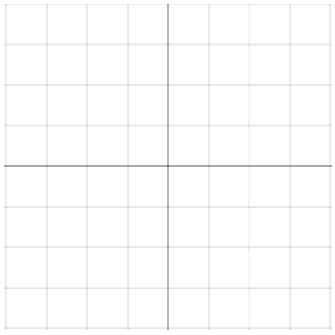
- **Given the equations below, sketch a graph after determining the following:**
  i. The x-intercept(s)
  ii. Behavior at the x-intercept(s)
  iii. Leading Coefficient sign
  iv. Degree of the Polynomial
  v. End Behavior

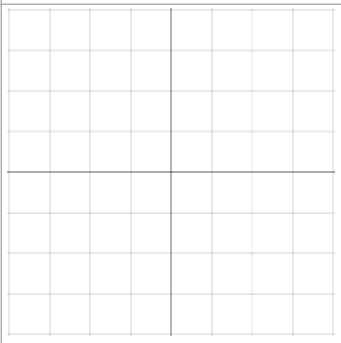
- $y = -\frac{1}{2}x(x+4)^2(x+1)^4(x-2)^3(x+2)$

x-intercept(s) & Behavior	*Sign of Lead Coefficient*	*Degree*	*End Behavior*	*Graph Sketch (don't scale the y-axis)*

- $y = -5x(x-1)^6(x+1)^7$

x-intercept(s) & Behavior	*Sign of Lead Coefficient*	*Degree*	*End Behavior*	*Graph Sketch (don't scale the y-axis)*

- $y = 4x^2(x-2)^2(x+1)^2$

x-intercept(s) & Behavior	*Sign of Lead Coefficient*	*Degree*	*End Behavior*	*Graph Sketch (don't scale the y-axis)*